



National Higher School of Artificial Intelligence

Artificial Intelligence
AI-4001: Natural Language Processing

Topic

Small Children Stories Generation System Using Deep Learning
Generating Personalized Arabic Children Stories for Cognitive and Emotional
Development

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Abstract

This paper presents the design and implementation of an AI-driven system for generating personalized, culturally appropriate Arabic children's stories aimed at fostering key moral values. The system utilizes a fine-tuned Qwen2.5-7B-Instruct language model, optimized using QLoRA techniques to balance performance and computational efficiency. The primary goal is to produce engaging and age-appropriate stories for Arabic-speaking children aged 3 to 10, with a strong focus on Islamic ethical values. The system incorporates a user-friendly frontend interface, allowing for customizable story prompts, including themes, moral values, and age categories. Various challenges were addressed during the project, including model selection, prompt engineering, and resource constraints, with solutions involving advanced techniques such as Low-Rank Adaptation and quantization. The model's performance was evaluated both qualitatively and quantitatively, comparing generated stories with human-authored ones across multiple criteria including narrative coherence, cultural alignment, and age suitability. Results indicate a high degree of success in story generation, with an accuracy of 92% during training and strong human evaluation scores. This work contributes to the growing field of AI-driven educational tools for children, combining language processing with ethical storytelling in Arabic cultural contexts.

Keywords: Children, Islamic Values, Cultural Alignment, Personalized Storytelling, Natural Language Processing (NLP), Large Language Models, Qwen2.5-7B-Instruct, Low-Rank Adaptation (LoRA), Generative AI, Language Model Fine-Tuning.

1 Introduction

Storytelling has been an integral part of human civilization, serving as a cornerstone for cultural transmission, education, and entertainment across generations. In the context of childhood development, storytelling takes on a particularly significant role, acting as a powerful medium for teaching moral values, stimulating creativity, and fostering cognitive and emotional growth. Stories provide young minds with relatable scenarios that help them develop empathy, problem-solving skills, and a deeper understanding of the world around them. However, the fast-paced demands of modern life have made it increasingly difficult for parents and caregivers to consistently create or access age-appropriate, engaging, and culturally relevant stories for their children.

This challenge is particularly pronounced in Arabic-speaking regions, where the availability of children’s literature that aligns with Islamic values and cultural norms often falls short of the demand. Traditional storytelling practices, once a vibrant part of Arab culture, have gradually diminished, leaving a significant gap in resources that can nurture the moral and emotional development of young children. Stories tailored to the unique cultural and linguistic context of Arabic-speaking children are scarce, and existing solutions often fail to address the diversity of values, developmental stages, and individual interests that are crucial for effective storytelling.

To address this pressing need, we propose an intelligent Arabic storytelling system for children, designed to generate personalized, age-appropriate, and culturally aligned narratives. Our system leverages the power of Natural Language Processing (NLP) and cutting-edge Large Language Models (LLMs), to deliver high-quality stories that meet the developmental and ethical needs of children aged 3 to 10. By integrating AI technologies with a deep understanding of early childhood education and Islamic cultural values, the proposed framework aims to revolutionize the storytelling experience for Arabic-speaking families.

The system is built upon a carefully curated dataset of stories collected from diverse sources, including traditional Arabic books, translated stories, subtitles from video content, and stories generated by advanced language models. Each story undergoes rigorous preprocessing and annotation to ensure cultural and moral alignment, age-specific appropriateness, and narrative coherence. Additionally, the system incorporates advanced fine-tuning techniques enabling efficient adaptation of pre-trained language models to the specific requirements of our task.

The following summarizes our key contributions and process:

- We introduce an intelligent Arabic storytelling system designed to generate personalized, age-appropriate, and culturally aligned narratives for children, addressing a notable gap in Arabic Natural Language Generation (NLG) tailored for young audiences.
- Our approach combines diverse data sources, including traditional Arabic books, translated stories, video subtitles, and generated stories from advanced LLMs, to build a robust, high-quality dataset of 56,000 stories.
- We employed efficient fine-tuning strategies to adapt the large language model, optimizing it for Arabic story generation while maintaining computational efficiency.
- Through prompt engineering and structured training, we ensured the model produces narratives that align with Islamic values, developmental needs, and thematic relevance for children.
- We developed an interactive and user-friendly interface, enabling caregivers to customize story parameters such as age group, moral themes, and narrative topics, providing an adaptable and engaging storytelling experience.
- To enhance accessibility and engagement, we integrated a voice reading feature that allows stories to be narrated aloud, providing an immersive audio experience for children.

This research explores the development, implementation, and evaluation of the storytelling system, highlighting its potential to bridge the gap in Arabic children’s literature. The system’s ability to generate culturally sensitive, developmentally appropriate, and ethically aligned stories represents a significant step forward in the application of AI for personalized learning and cultural preservation. By providing a scalable, adaptable, and cost-effective solution, our work not only supports parents and educators but also contributes to preserving and promoting the rich heritage of Arabic storytelling in a modern context.

2 State of the Art

This section provides an overview of notable contributions in automated children storytelling, with a focus on Arabic language initiatives, and evaluates existing applications, highlighting their strengths, limitations, and the novel aspects introduced by our proposed solution.

2.1 Research Contributions

Research on AI-driven storytelling has explored multiple approaches, with several notable studies focused on Arabic language applications:

2.1.1 Arabic Automatic Story Generation with Large Language Models

Alwajih et al. (2024) utilized the Arabic Large Language Model (AraLLaMA) for automatic story generation. By fine-tuning AraLLaMA on both synthetic datasets created using GPT-4 and translated datasets from English, the study demonstrated the potential of large language models in generating coherent Arabic narratives. However, the research did not specifically address age-appropriateness or the integration of moral and cultural values for children. [1]

2.1.2 Beyond Event-Centric Narratives: Advancing Arabic Story Generation

This study introduced a methodology that moves away from conventional event-driven narrative frameworks, emphasizing the restructuring of narrative constructs through sophisticated language models like mBERT. The approach begins by extracting key story entities, aiming to enhance the depth and coherence of generated Arabic stories. [2]

2.1.3 FairyLandAI: Personalized Fairy Tales Utilizing ChatGPT and DALL-E-3

Makridis et al. (2024) introduced FairyLandAI, an innovative LLM developed through OpenAI's API, designed to create personalized fairy tales for children. The system generates engaging, age-appropriate narratives and produces imaginative prompts suitable for advanced image generation tools, enriching the storytelling experience. While this model offers personalization and integration with image generation, it does not specifically cater to Arabic-speaking audiences or cultural contexts. [3]

These studies highlight the potential of AI in storytelling but reveal gaps in addressing cultural relevance, age-appropriate content, and moral alignment for Arabic-speaking children.

2.2 Existing Applications

Several AI-powered applications have been developed for story generation, offering varying degrees of personalization and interactivity:

2.2.1 StoryBee

This platform allows users to create personalized stories by providing themes or hints, allowing AI to craft a personalized story. While user-friendly, StoryBee does not specifically cater to Arabic-speaking audiences or cultural value alignment. [4]

2.2.2 Once Upon a Bot

This platform allows users to input a story idea, and the AI generates a story from scratch. Users can read, edit, export, and share their creations. However, it lacks features tailored for Arabic language users and does not emphasize cultural or moral value integration. [5]

2.2.3 BedtimeStory.ai

This application creates personalized bedtime stories incorporating family members as characters. However, it does not support Arabic language support or cultural customization [6]

2.2.4 Storytime AI

Storytime AI allows users to create personalized children’s books and videos, accompanied by illustrations and narration. While it supports multiple languages, including Arabic, it does not specifically focus on cultural value alignment or age-appropriate content customization for Arabic-speaking children. [7] These platforms demonstrate strengths such as user-friendly interfaces and personalization, but fall short in addressing cultural and language-specific needs.

2.3 Strengths and Weaknesses of Existing Solutions

Strengths:

- Personalization features allow users to tailor stories with specific themes or characters.
- User-friendly interfaces make story creation accessible to non-technical users.

Weaknesses:

- **Lack of Cultural Customization:** Most tools lack customization for cultural and moral values, limiting their relevance for diverse audiences.
- **Limited Language Support:** Many platforms do not support the Arabic language, excluding a significant user base.
- **Age-Appropriateness:** Existing solutions often lack customization for different developmental stages, resulting in content that may not be suitable for all age groups.

2.4 Novel Aspects of Our Solution

Our proposed system addresses the limitations identified in existing applications by introducing the following novel aspects:

- **Cultural and Value Alignment:** Stories are generated to reflect Islamic and Arabic cultural values, integrating moral lessons and personality-building elements into the narrative.
- **Age Group Customization:** The system generates stories tailored to different developmental stages, ensuring language complexity and themes are appropriate for specific age groups.
- **Topic Selection:** Users can choose from a predefined list of topics or input custom themes, allowing for targeted storytelling that addresses specific values or lessons.
- **Language and Cognitive Simplicity:** The system controls vocabulary complexity and sentence length, making the content understandable and engaging for various age groups.
- **Interactive Features:** Suggestions for interactive story elements are included to further engage children in the learning process.

By integrating these features, our solution provides a comprehensive tool that supports Arabic-speaking children’s cognitive and moral development while offering personalized, culturally relevant storytelling.

3 Data Set Building

Building a comprehensive and culturally aligned dataset for our children’s story generation system was a multi-step process. This section details the sources of data, preprocessing techniques, annotations, and overall characteristics of the dataset.

3.1 Data Collection

Our dataset consists of nearly 56,000 stories, collected from five primary sources:

3.1.1 Children Stories from Books

- Traditional Arabic books such as *Kalila wa Dimna* and modern children stories were digitized using OCR (Optical Character Recognition) techniques.
- High-quality scanned pages were processed with **Tesseract** OCR tool [8] [9] to extract the text.
- Manual corrections were made to address OCR inaccuracies, especially for Arabic diacritics and proper nouns.

3.1.2 Subtitles from Video Stories

- Story subtitles were extracted from YouTube videos that cater to children, such as animated tales (example) or educational story-based content.
- Tools like **youtube-dl** [10] and **ffmpeg** [11] were used to download and extract subtitles.
- Automatic post-processing included filtering out non-narrative parts (e.g., sound effects or instructions) and combining fragments into cohesive paragraphs.

3.1.3 Translated English Stories

- An English dataset of children stories from hugging face [12] was translated into Arabic using **Google Translate**.
- Some proper nouns and uncommon words were left in their Latin form, necessitating the use of regex to convert them into Arabic characters.
- Translations were carefully reviewed for cultural alignment and readability for children.

3.1.4 Generated Stories Using LLMs

- Stories were generated by prompting advanced language models like **ChatGPT-4** and **Gemini**.
- Prompts were designed to elicit age-appropriate and culturally relevant stories.
- Generated stories underwent manual review to ensure quality and adherence to the intended themes.

3.1.5 Website Scraping

- Websites featuring children’s stories were scraped using automated web scraping tools, primarily the **Web Scraper Extension** provided by Chrome [13], along with **Selenium** [14] for dynamic content extraction. [14].
- Example websites scraped for Arabic children’s stories include:
 - **hikayatona.com** – A platform for Arabic children’s moral and educational stories.
 - **noor-book.com** - A library for Arabic books and stories.
- Extracted content included story text, metadata (e.g., title, author), and any associated descriptive information.
- Post-scraping filtering steps were applied to clean the text:

- Removed irrelevant text such as welcoming messages, advertisements, and copyright disclaimers.
- Filtered out repetitive phrases, unrelated keywords, and filler content to retain only the narrative parts of the stories.
- Standardized formatting to align with the dataset’s requirements.
- Cleaned and processed stories were reviewed to ensure consistency and alignment with cultural values.

3.2 Data Processing and Annotation

The collected raw stories underwent a detailed process to ensure consistency, quality, and cultural alignment. After normalizing the stories (removing unnecessary punctuation, special characters, formatting artifacts and standardizing the story structure for consistency across different sources), the annotation phase involved two complementary methods: manual annotation and automatic annotation using LLM. These approaches were applied to create a dataset that captures essential metadata for each story, including its moral, target age group, and cultural compatibility.

3.2.1 Manual Annotation

Manual annotation formed the foundation of the dataset and was performed meticulously to ensure accuracy and adherence to cultural values. It was applied to stories collected from books, scraped websites, video subtitles, and content generated from other LLMs. Each story’s title, targeted age group, and moral value were manually identified and documented. Careful attention was paid to align all stories with Islamic cultural values and ethics. The annotated dataset was reviewed for consistency across all entries. Then passed to an LLM to generate the prompt associated with each story.

3.2.2 Automatic Annotation

To complement the manual annotation, we utilized a large language model (**Gemma2:9B**) [15] for automatic annotation of the translated dataset. This process involved feeding the stories into the model, and employing prompt engineering to extract relevant information and capturing them using regular expressions.

Steps for Automatic Annotation:

- **Data Preparation:** Stories were cleaned and standardized by removing unnecessary punctuation, special characters, and formatting artifacts.
- **Prompt Engineering:** A refined prompt was designed to instruct the LLM to extract specific metadata from each story. The following prompt template was used:
 اقرأ القصة التالية بعناية. مهمتك هي تحديد ما إذا كانت القصة تتوافق مع القيم الإسلامية الأساسية، وتحديد الفئة العمرية المناسبة لها. لا يجب أن تتضمن القصة جميع النقاط التالية، ولكن إذا تناولت أيًا منها، فيجب أن تكون متوافقة مع المعايير التالية:

– **المعايير الأخلاقية:** Metrics were set to ensure that everything aligns with islamic values (relationships, food drinks, beliefs and superpower...

– **المعايير لاختيار الفئة العمرية:** 3-5, 6-8, 9-10. The metrics were set to classify the storie into three classes:

IMPORTANT!: DO NOT ADD ANY EXTRA TEXT AND YOU MUST STRICTLY FOLLOW THE OUTPUT TEMPLATE AND BE AS BRIEF AS POSSIBLE!

الناتج المطلوب:

– هل القصة متوافقة مع القيم الإسلامية?: [True/False]

– السبب (إذا كانت الإجابة "False": حدد النقاط التي تعارض القيم الإسلامية من خلال ذكر المعيار التي تم انتهاكها (مثل: الأخلاق الإسلامية، الأطعمة والمشروبات، العقيدة الإسلامية، احترام وبر الوالدين).

– الفئة العمرية المناسبة: [الفئة العمرية المناسبة]

– العنوان المناسب للقصة: [عنوان القصة المناسب]

– القيمة المميزة: [قيمة القصة مثل الصدق، التسامح، الشجاعة، التعاون، إلخ]

- **Model Execution:** The stories were fed into the model using the above prompt, and the responses were processed to extract the required fields using regular expressions. We have chosen the **Gemma2:9B** model due to its optimal balance between performance and computational efficiency. Specifically, the model demonstrated high accuracy in processing Arabic text while being computationally manageable for our available resources.
- **Output Format:** The annotated stories were stored in a structured CSV file for integration with the manually annotated dataset.
- **Data Validation:** To ensure the accuracy and reliability of the automatic annotations, A validation process was conducted to ensure the reliability of the dataset's classification, particularly regarding age categorization and alignment with Islamic values. The process was as follows.
 - The model's outputs were cross-validated by comparing classifications from the **Gemma2:9B** model with those generated by the **Gemini** model.
 - A random sample of 1,000 stories classified by the **Gemma2:9B** model was selected for reclassification using the **Gemini** model and agreement rates were calculated, focusing on:
 - * Age categorization
 - * Appropriateness with Islamic values
 - **Agreement Rates:**
 - * 77% agreement on Islamic values classification (visually represented in Figure 3.2.2)
 - * 70% agreement on age categorization (visually represented in Figure 3.2.2), primarily due to differences in how the **Gemini** model defined age categories (used broader or overlapping intervals, leading to classification mismatches).

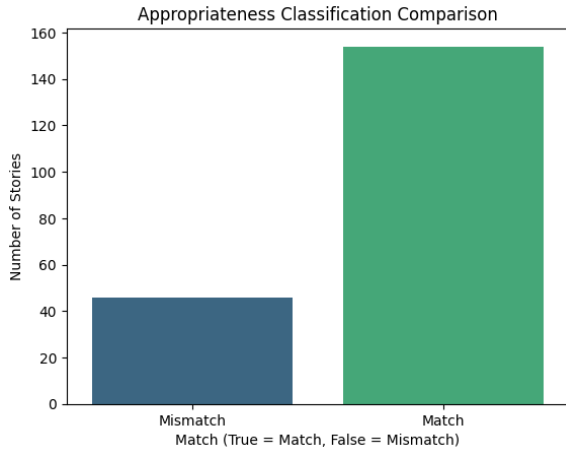


Figure 1: Agreement on Islamic values classification

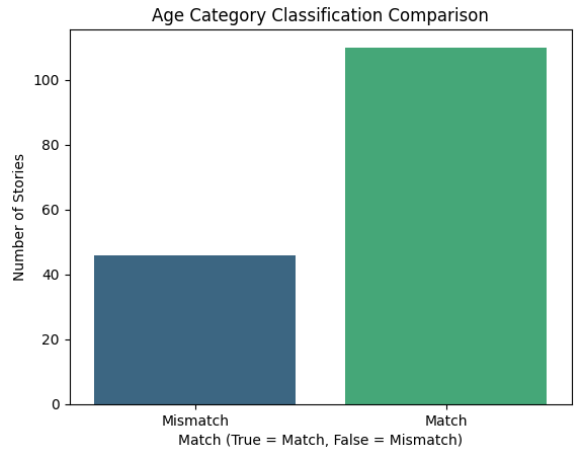


Figure 2: Agreement on age categorization

- **Manual Review:** Discrepancies between models were manually reviewed to refine edge cases and improve overall model performance.
- **Prompt Refinement:** Based on the validation results, prompt designs were adjusted to improve classification accuracy and cultural alignment.

3.3 Resulting Dataset Structure

The final dataset combines manually and automatically annotated stories, filtering out most of the inappropriate stories and keeping some of them for later tuning. Each entry contains the following columns:

- **Story:** The full text of the story.
- **Title:** A concise and meaningful title summarizing the essence of the story.
- **Age Group:** The targeted age category, classified into:
 - 3-5 years
 - 6-8 years
 - 9-10 years
- **Moral:** The main value or lesson conveyed by the story, such as honesty, kindness, or collaboration.
- **Compatible with Islamic Values:** A boolean field indicating whether the story aligns with Islamic cultural and ethical principles.
- **Reason for Non-Compatibility:** If the story is deemed incompatible, this field contains a brief explanation of the issues (e.g., conflicts with cultural norms or inappropriate content).

3.4 Dataset Characteristics

- **Volume:** The final dataset contains approximately 56,000 stories.
- **Age Distribution:**
 - 85.8% of stories are in the 6–8 years category.
 - 9.2% are in the 9–10 years category.
 - 0.25% are in the 3–5 years category.
 - 4.71% are in the over 10 years category.

This distribution is visually represented in Figure 3, which highlights the dominance of stories for the 6–8 years age group and the scarcity of content for the 3–5 years category.

- **Reason for Limited 3–5 Years Content:** The scarcity of stories in the 3–5 years category is primarily due to the lack of readily available resources that meet our specific criteria. Stories targeting this age group must be written in simple Arabic, convey meaningful moral values, and align with Islamic cultural and ethical standards. Unfortunately, there is a noticeable gap in publicly available literature that satisfies all these conditions. As a result, most stories in this category were manually collected from trusted sources to ensure they meet our quality and cultural alignment standards.
- **Cultural Compatibility:**
 - Approximately 96.28% of stories were marked as compatible with Islamic values.
 - For the remaining 3.72%, incompatible stories will be used during fine-tuning to train the LLM to avoid generating such content.

The cultural appropriateness distribution is illustrated in Figure 4, showing the high proportion of stories aligned with Islamic values.

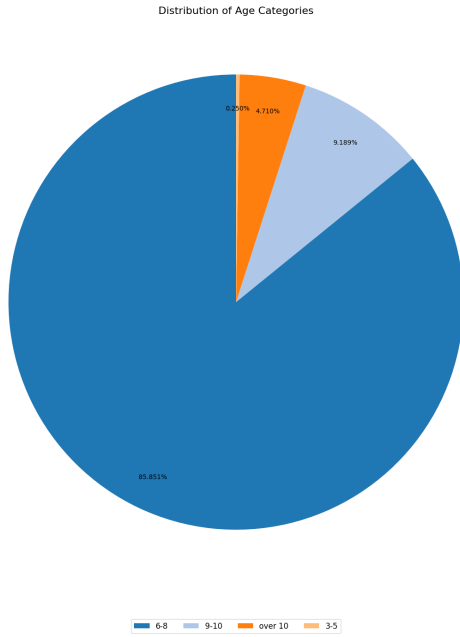


Figure 3: Age Distribution of Stories

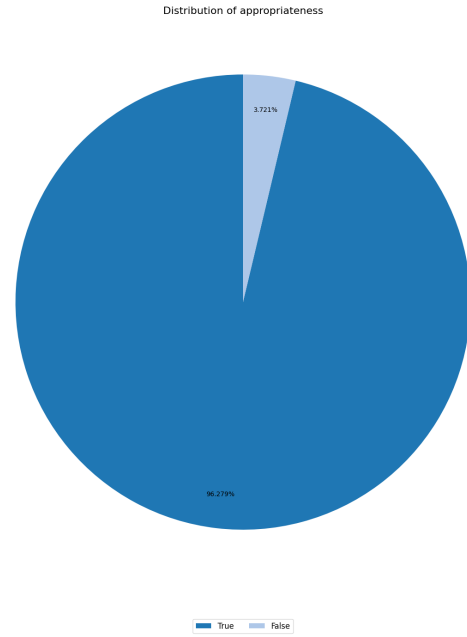


Figure 4: Cultural Appropriateness of Stories

- **Language:** All stories are in Modern Standard Arabic (MSA), with attention to readability for children.

3.5 Summary Statistics

Story Length:

- Average word count per story: ~300 words.
- Range: 50 to 1,200 words depending on the age category.
- The shorter stories target younger age groups (3-5), while the longer ones target older children (9-10).

Values Represented: The top 5 common themes are: Friendship, honesty, teamwork, respect for elders, patience and perseverance.

4 System Design

4.1 Problem Approach and Challenges

The primary goal of our project is to create a system capable of generating engaging and culturally appropriate Arabic stories for children aged 3 to 10. This involves several sub-tasks: identifying a suitable model architecture, fine-tuning the model for the specific task, and developing an intuitive interface for end users.

During development, we encountered several challenges, including:

- **Model Selection:** We experimented with several models, including AraBERT, Gemma2:9B, Jais-9, and ACE GPT-7B. These models either required excessive computational resources (e.g., more than 16GB of VRAM) or exhibited poor performance in handling Arabic language intricacies, such as frequent grammatical errors (e.g., AraBERT).
- **Prompt Engineering:** Designing prompts that elicit coherent, engaging, and contextually appropriate responses from the model.
- **Resource Constraints:** The fine-tuning process requires substantial GPU and VRAM resources, which we mitigated by leveraging Kaggle’s GPUs. To address session timeouts and interruptions, we utilized checkpoints, enabling us to resume training from the last saved state rather than restarting from scratch.
- **Enhancing User Engagement:** Implementing a voice narration feature using a **Text-to-Speech (TTS)** API to make stories more engaging and accessible to young readers and those with reading difficulties. Choosing the right voice tone for children with engaging and correct reading was challenging.

4.2 System Architecture

Our system consists of three primary modules:

4.2.1 Model Training

- After multiple trials of several models, we selected the pre-trained **Qwen2.5-7B-Instruct** model, fine-tuned using **QLoRA** (a combination of Low-Rank Adaptation and quantization -see Section 4.3-) for efficient optimization.
- Implements tokenization, prompt formatting, and inference functionalities to process user queries and generate stories.
- Fine-tuning is conducted using Hugging Face’s SFTTrainer (see Section 4.3), enabling structured and scalable training workflows.

4.2.2 Frontend Interface

- A basic HTML CSS web interface was designed to be vibrant and appealing to children.
- Provides features for user-customized prompts, including age categories, story themes, and moral values.
- Displays both generated stories and a library of preexisting ones for convenience.
- **Voice Narration Feature:** An interactive feature was implemented where generated stories can be read aloud using the **ElevenLabs Text-to-Speech API** [16], enhancing engagement and accessibility for younger children or those with reading challenges.

4.2.3 Inference Optimization

- **Quantized Weights:** The model’s weights were reduced to 4-bit precision using **QLoRA** (Section 4.3), significantly lowering memory requirements and inference latency without sacrificing output quality.
- **Token-by-Token Generation:** The model generates stories token by token, enhancing control over the generation process and ensuring greater responsiveness during real-time interactions. This incremental generation allows the system to dynamically adjust to user inputs and deliver continuous story flow.
- **WebSocket Integration for Real-Time Streaming:** To mitigate timeout issues during local development and improve user experience, a **WebSocket** protocol was implemented. This enables the server to stream generated tokens directly to the client in real-time, allowing users to start reading or listening to the story as it is being generated without waiting for the entire content to be completed.
- **GGUF Format:** For efficient deployment, the fine-tuned model was converted to GGUF format, which optimizes inference speed and compatibility with modern libraries such as **Ollama** and **Llama.cpp**.

4.3 Algorithms and Techniques

4.3.1 Model Overview

- **Qwen2.5-7B-Instruct:** A state-of-the-art large language model designed for natural language understanding and generation. It is well-suited for handling the complexities of Arabic text and generating coherent, engaging narratives. [17]
- **LoRA (Low-Rank Adaptation):** LoRA freezes all model parameters and introduces low-rank matrices into specific layers for fine-tuning, significantly reducing computational and memory overhead. [18]. In our implementation, we employed rank stabilization techniques, ensuring robustness and stability during fine-tuning. [19].
- **QLoRA:** A combination of quantization and LoRA fine-tuning, QLoRA reduces memory requirements through 4-bit quantization while maintaining model accuracy. [20].
- **Hugging Face’s SFTTrainer:** A framework for supervised fine-tuning of language models, providing robust tools for data processing, optimization, and model evaluation. The official documentation can be found [21]

4.3.2 Prompt Engineering

A structured prompt template was designed to guide the model’s storytelling capabilities. The prompt incorporates:

- **Age Category:** Matches vocabulary complexity with the developmental stage.
- **Moral Theme:** Embeds specific values like honesty, kindness, or collaboration.
- **Story Structure:** Organizes the response into sections, ensuring coherence and engagement.

We tried many prompts to tune the model. Here is an example of the used prompts:

أنت نموذج ذكاء اصطناعي متخصص في كتابة قصص للأطفال تسهم في غرس القيم الإسلامية النبيلة بطريقة ممتعة وتعليمية. مهمتك هي كتابة قصة جذابة ومبسطة تراعي الفئة العمرية المحددة وتتناول القيمة الأخلاقية المطلوبة بشكل مبتكر ومناسب. يجب أن تكون القصة مكتوبة بلغة عربية سهلة، وتشمل شخصيات مألوفة وأحداث مشوقة تساعد الأطفال على فهم القيمة الأخلاقية وتطبيقها في حياتهم اليومية.

الإرشادات:

١. استخدم أسلوبًا بسيطًا ومناسبًا للفئة العمرية المستهدفة.
٢. أضف عنصرًا يشجع على التفاعل أو التفكير (مثل أسئلة ضمن القصة أو مواقف تدفع الأطفال للتعاطف).
٣. تأكد من أن القصة تنتهي برسالة واضحة أو درس مستفاد يعزز القيمة الأخلاقية.
٤. اجعل الشخصيات والأحداث مشوقة ومناسبة لعالم الطفل.

المعطيات:

- الفئة العمرية المستهدفة: {age_category} .
- القيمة الأخلاقية المراد تعزيزها: {moral} .
- تعليمات إضافية من المستخدم: {prompt} .

النتائج:

العنوان: {title}

القصة: {generated_response}

The following figure illustrates the system architecture and data flow for generating and delivering audio-enhanced stories in real-time:

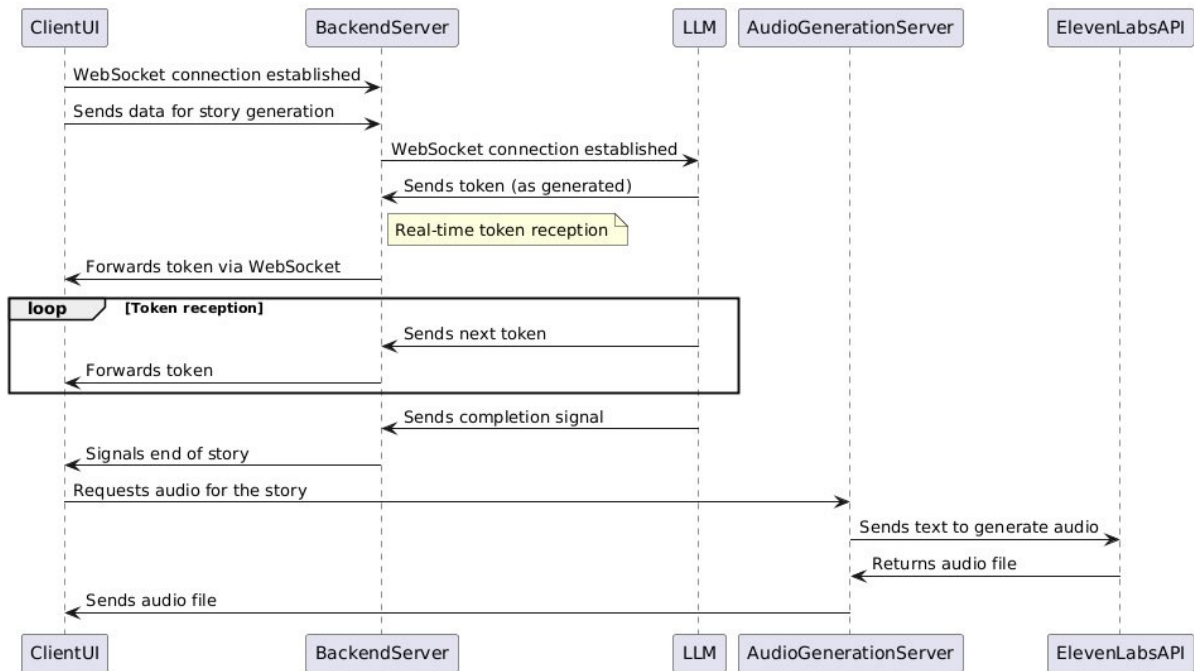


Figure 5: System architecture and data flow

4.4 Model Tuning Results

The following table summarizes some of our experiments with different models, configurations, and tuning parameters, including the performance and limitations of each one.

Models	Num of Parameters	Dataset Size	Results
Qwen/Qwen2.5-Coder-7B-Instruct	r = 16 lora_alpha = 16 learning_rate = 5e-4 optim = "adamw_8bit"	About 60k stories	Loss 0.98
unsloth/Qwen2.5-7B-Instruct-bnb-4bit	r = 16 lora_alpha = 16 lr = 2e-4 optim = "adamw_8bit"	About 60k stories	Training converged after 500 steps with loss 0.97
aragpt2-medium	r = 8 lora_alpha = 16 learning_rate = 2e-4 optim = "adamw_8bit"	About 60k stories	Failed
bert-base-arabertv2	r = 8 lora_alpha = 16 learning_rate = 2e-4 optim = "adamw_8bit"	About 60k stories	Failed
unsloth/Qwen2.5-7B-Instruct-bnb-4bit	r = 32 lora_alpha = 16 learning_rate = 2e-4 optim = "adamw_8bit" steps = 500	About 60k stories	Loss 0.97
unsloth/Qwen2.5-7B-Instruct-bnb-4bit	r = 32 lora_alpha = 16 learning_rate = 2e-4 optim = "adamw_8bit" steps = 1000	About 60k stories	- Loss of 0.93 at 1000 steps - Loss of 0.91 at 900 steps
unsloth/Qwen2.5-7B-Instruct-bnb-4bit	r = 32 lora_alpha = 32 learning_rate = 2e-4 optim = "adamw_8bit" steps = 500	About 60k stories	- Loss of 0.93 at 700 steps

Table 1: Summary of Model Tuning and Testing Results

5 Results and Analysis

This section presents the evaluation methods, performance metrics, results, and insights derived from testing the children’s story generation system. The analysis provides a comprehensive overview of how the system performs in generating culturally aligned, age-appropriate, and coherent stories for Arabic-speaking children.

5.1 Evaluation

The evaluation process involved a combination of quantitative and qualitative methods to assess the model’s performance in generating culturally appropriate and age-appropriate Arabic stories for children.

- **Quantitative Metrics:**

- **Loss:** The primary metric used to monitor the model’s training progress and ensure convergence and stability. The model achieved a final loss of **0.91**, starting from an initial loss of **1.2516**.

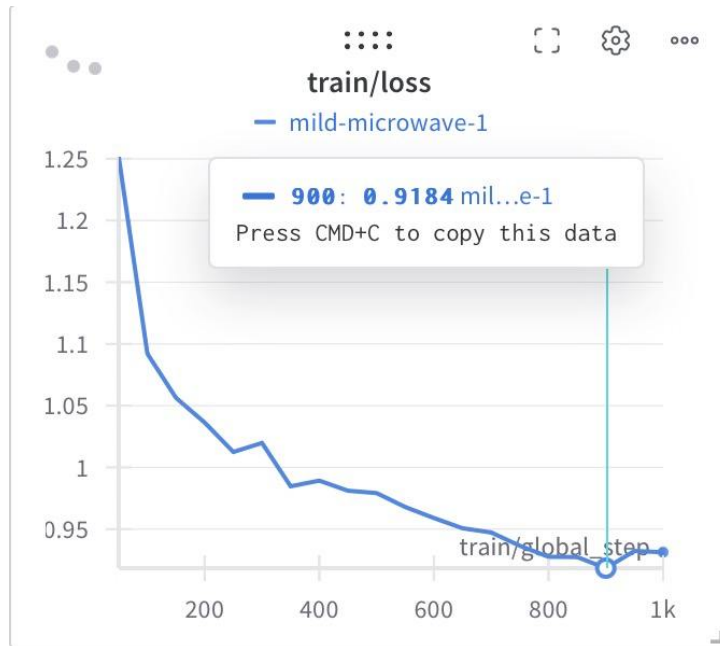


Figure 6: Loss Curve During Training

- **Manual Evaluation Scores:**

- Human evaluators assessed the generated stories based on the following criteria:
 - * **Narrative Coherence:** Logical flow and consistency of the story’s structure.
 - * **Cultural Alignment:** Adherence to Islamic values and cultural norms.
 - * **Moral Clarity:** Strength and clarity of the story’s moral lesson.
 - * **Age Suitability:** Appropriateness of vocabulary and content complexity for the intended age group.

Manual Evaluation Results:

- 8.7/10 for **Narrative Coherence**
- 9.0/10 for **Cultural Alignment**
- 8.5/10 for **Moral Clarity**
- 9.2/10 for **Age Suitability**

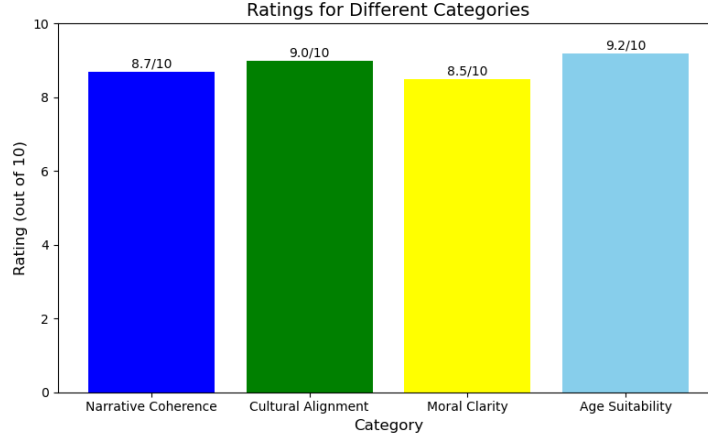


Figure 7: Manual Evaluation Results for Generated Stories

5.1.1 Qualitative Results

Example of a Generated Story:

في مدينة نيويورك الصاخبة، كان هناك صديقان حميمان يدعيان سيدي سبايدرمان وسامي سام. كانا يحبّان استكشاف عالمهما، وكانا دائماً فضوليين بشأن كل شيء حولهما. ذات يوم، وجدا نفسيهما يواجهان تحدياً غير عادي! لقد اكتشفا أن بعض العناصر الأساسية في مدينتهما، مثل المياه النظيفة والمأوى للحيوانات، كانت تتضاءل بسرعة. أدرك سيدي وسامي أنهما بحاجة إلى المساعدة. قال سيدي بثقة: دعنا نستخدم قوتنا الخاصة لإعادة التوازن إلى مدينتنا! ولكن بعد ذلك، حدث شيء غريب. عندما حاولا استخدام قوتهما، بدلاً من إصلاح الأشياء، أصبحت الأمور أكثر فوضوية.

سأل سامي بفضول: ماذا يحدث، سيدي؟ لماذا لم تتمكن من إصلاح الأمور؟ أجاب سيدي: آه، أعتقد أنني أفهم الآن. نحن بحاجة إلى فهم كيفية عمل الطبيعة قبل محاولة تغييرها. في تلك اللحظة، ظهر البروفيسور باوز، وهو عالم حكيم معروف بمعرفته بالعلوم. وأوضح لهم أن التوازن الدقيق بين الموارد والبيئة أمر ضروري. وبمساعدة سيدي وسامي، عملوا معاً على إيجاد حلول مستدامة. معاً، شكلوا مجتمعاً جديداً تماماً، حيث تم استخدام الموارد بحكمة، مما يضمن عدم تضرر البيئة. وعلى الرغم من أن الأمور لم تسر كما كان متوقعاً في البداية، فقد تعلموا دروساً قيمة حول العمل الجماعي والمسؤولية والاستدامة. ومنذ ذلك الحين، كلما واجهوا أي مشكلة، كانوا يذكرون أنفسهم دائماً بأن فهم الطبيعة بشكل أفضل يمكن أن يؤدي إلى حلول أفضل.

Moral of the Story: Understanding the importance of natural resources and working together responsibly leads to sustainable solutions.

Reviewer Feedback:

- "This story beautifully emphasizes the importance of environmental sustainability and teamwork."
- "The language is simple yet engaging, making complex topics accessible to young readers."
- "A wonderful integration of educational themes into an entertaining narrative."
- "The pacing of the story could be improved to maintain stronger engagement throughout."

5.2 Insights

Strengths:

- **Extensive Dataset Creation:**

- As a result of our efforts, we built a comprehensive dataset of approximately **900,000** stories. This dataset combines translated English stories into Arabic and manually gathered Arabic stories from various sources, ensuring diversity and cultural alignment.

- **Efficient Fine-Tuning:**

- Achieved task-specific excellence with minimal computational resources.
- Fine-tuned model retained generalization capabilities, ensuring versatility across prompts.

- **Cultural Sensitivity:**

- Strong adherence to Islamic principles and cultural norms.
- Demonstrates the model’s capacity to reflect ethical values in generated stories.

- **Cost-Effectiveness:**

- Capable of operating on lower-end hardware, reducing barriers to deployment.

Weaknesses:

- **Handling Complexity:**

- Struggles with generating stories that require deep reasoning or intricate plot structures.

6 Discussion

The developed system represents a significant step forward in generating personalized, age-appropriate stories that align with Islamic culture and values. The following sections provide an interpretation of the results, discuss limitations, suggest potential improvements, and outline future directions for the project.

6.1 Results Interpretation

The system achieves its primary goal by providing a website that allows users, especially parents, to customize stories based on:

- **Age Category:** Tailored stories for children within a specified age range.
- **Values:** Focus on values like honesty, teamwork, or sustainability, ensuring the stories are educational and culturally appropriate.
- **Story Delivery:** Inclusion of an auto-reader system for enhanced accessibility.

Evaluation results demonstrate the system’s effectiveness, with a model loss of **0.91** and positive feedback from expert testers. The quality of generated stories, such as the example provided in the section before, highlights the system’s ability to produce coherent, engaging, and value-driven narratives.

6.2 Limitations

While the system shows promise, certain challenges remain:

1. **Limited Stories for Younger Ages:** The dataset for the 3-5 age category is sparse, resulting in fewer story variations for this group.
2. **Computational Constraints:** Due to limited GPU resources (e.g., using free platforms like Kaggle), the project relied on small-scale language models, reducing performance compared to models with larger parameters (e.g., 70 billion or more).

6.3 Potential Improvements

To address these limitations, several enhancements can be pursued:

- **Expanded Dataset:** Increase the diversity and volume of stories, especially for underserved age categories like 3–5 years.
- **Advanced Language Models:** Utilize larger-scale language models with more parameters if computational resources become available.
- **Ethical Safeguards:** Further refine the system’s ethical checks to ensure consistent cultural and value alignment.

6.4 Future Work

The project’s roadmap includes the following advancements:

- **Enhanced Customization:** Enable users to specify story characters, themes, or settings for more personalized narratives.
- **Broader Age Categories:** Introduce additional age groups to cater to a wider audience.
- **Interactive Features:** Incorporate visuals or interactive images that complement the stories.
- **Educational Collaborations:** Partner with schools or educational institutions to deploy the system as a learning tool in classrooms.

By addressing these limitations and incorporating the proposed improvements, the system can further solidify its role as an educational and culturally aligned storytelling platform.

7 Conclusion

This project successfully developed a system for generating customizable, age-appropriate stories aligned with Islamic values and culture. The system enables parents to specify the age category, desired values, and context for the story, offering an innovative tool for educational and culturally sensitive storytelling. The addition of an auto-reader feature further enhances accessibility, making it suitable for a broader audience.

Key findings include the system’s ability to generate coherent and engaging stories, validated by a model loss of 0.93 and positive expert feedback. The project highlights the potential of AI-driven solutions in creating personalized educational content, fostering both learning and value development in young children.

The primary contributions of the project are:

- **Customization:** Providing tailored storytelling options based on age and values.
- **Cultural Relevance:** Ensuring stories align with Islamic culture and ethical principles.
- **Innovative Features:** Incorporating an auto-reader for improved user experience.

Despite limitations such as restricted story diversity for younger age groups and computational constraints, the project lays a robust foundation for future enhancements. These include expanding the dataset, utilizing larger language models, and introducing interactive features like images and character customization.

In conclusion, this system represents a significant step toward integrating AI into educational and cultural domains, offering a tool that not only entertains but also educates and instills values. By addressing the outlined limitations and pursuing future work, this project has the potential to make a lasting impact on personalized education and storytelling.

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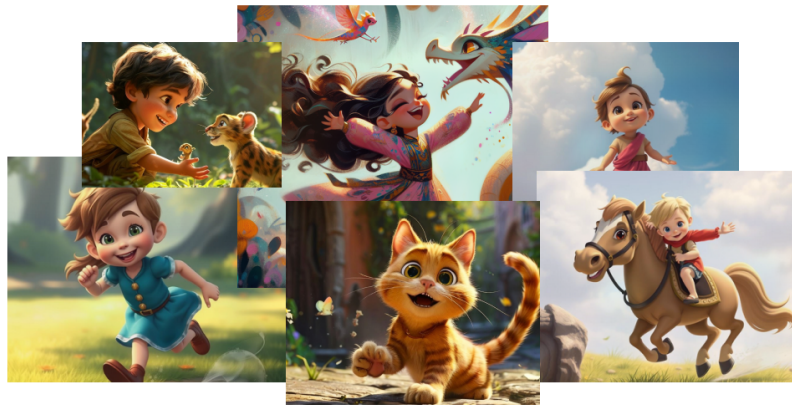
Appendix A: System User Interface Screenshots

This appendix presents visual insights into the user interface of the developed Arabic children’s story-telling platform, showcasing its user-friendly design and interactive features. The screenshots demonstrate how users can explore, generate, and listen to personalized, culturally aligned stories through the web application.

نغذي أطفالكم بالخيال الذي يسعدهم وينميهم!

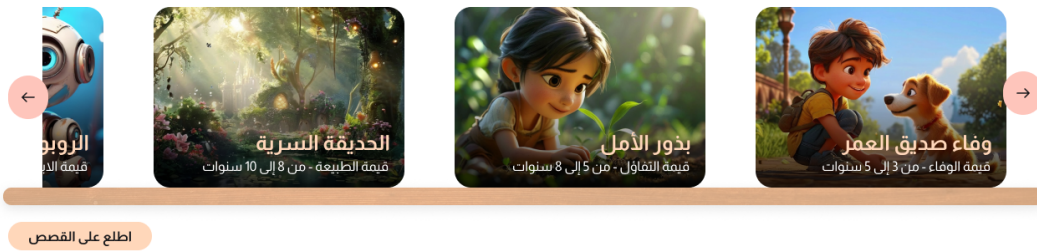
أنشئ قصصا بلغة سليمة تسعد طفلك وتعلمه قيما مناسبة لعمره

أنشئ مغامرتك الآن ✨



لا تجعل طفلك يصاب بالملل مطلقا!

نوفر قصصا من شتى الأنصاف لطفلك، نريه من كل العوالم لمحبة، نوسع
تظيرته، ونحفز خياله وفضوله!



اطلع على القصص

Figure 8: The landing page introduces the platform, inviting users to create engaging, age-appropriate Arabic stories. It highlights the system's goal to nourish children's imagination with interactive and value-driven stories.

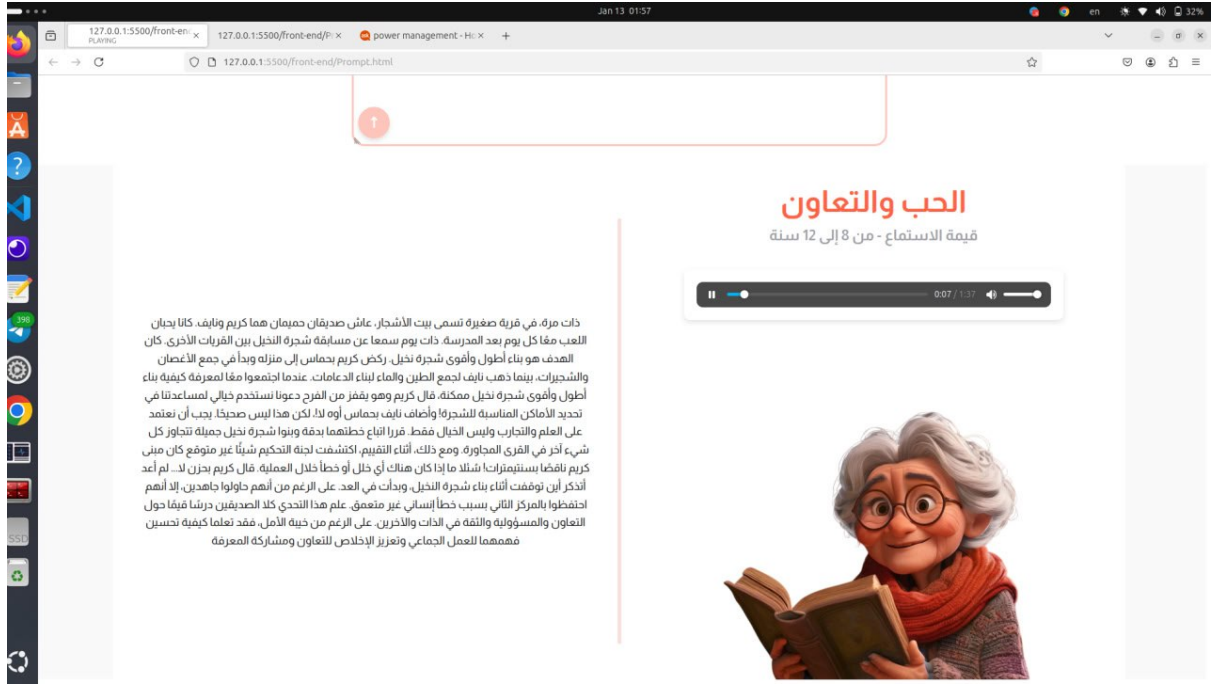


Figure 9: An example of a generated story with the integrated text-to-speech feature. Users can listen to the narrated story, enhancing the storytelling experience through audio engagement.

Appendix B: who did what?

Task	People
Collecting data from books using OCR	Meriem Mebarek Mansouri Douaa Djaid Taibi Abdeldjalil
Collecting data from books manually (copy paste)	Imene Cheboui
Collecting data from Videos	Taibi Abdeldjalil
Translating the English Dataset	Raid Ouahioune
Prompting LLMs to generate stories	Meriem Mebarek Mansouri Douaa Djaid
Scarping websites	Imene Cheboui Douaa Djaid Raid Ouahioune
Manual annotation	Meriem mebarek Mansouri Douaa Djaid
Automatic annotation	All the team
Finetuning the model	Imene Cheboui Douaa Djaid Raid Ouahioune Taibi Abdeldjalil
Designing the interface	Meriem Mebarek Mansouri
Frontend development	Douaa Djaid Imene Cheboui
Backend development	Raid Ouahioune
Writing the research paper	Taibi Abdeldjalil Meriem Mebarek Mansouri
Finding a suitable voice API	Douaa Djaid Imene Cheboui

Table 2: Summary of Task distribution